Quantitative Aptitude — Complete Preparation Guide

For CSE/SDE Placement Tests (TCS NQT, Infosys, Wipro, Capgemini, etc.)

Study Notes

Contents

1 How to Use This Guide **3**

2 1. Number System **4**
2.1 1.1 Key Concepts & Formulas . . . 4
2.2 1.2 Important Notes / Shortcuts . . . 4
2.3 1.3 Solved Examples . . . 4
2.4 1.4 Practice MCQs . . . 4

3 2. HCF and LCM **6**
3.1 2.1 Key Concepts & Formulas . . . 6
3.2 2.2 Important Notes / Shortcuts . . . 6
3.3 2.3 Solved Examples . . . 6
3.4 2.4 Practice MCQs . . . 6

4 3. Percentages **7**
4.1 3.1 Key Concepts & Formulas . . . 7
4.2 3.2 Important Notes / Shortcuts . . . 7
4.3 3.3 Solved Examples . . . 7
4.4 3.4 Practice MCQs . . . 7

5 4. Profit, Loss and Discount **8**
5.1 4.1 Key Concepts & Formulas . . . 8
5.2 4.2 Important Notes / Shortcuts . . . 8
5.3 4.3 Solved Examples . . . 8
5.4 4.4 Practice MCQs . . . 8

6 5. Simple Interest and Compound Interest **9**
6.1 5.1 Key Concepts & Formulas . . . 9
6.2 5.2 Important Notes / Shortcuts . . . 9
6.3 5.3 Solved Examples . . . 9
6.4 5.4 Practice MCQs . . . 9

7 6. Ratio, Proportion and Averages **10**
7.1 6.1 Key Concepts & Formulas . . . 10
7.2 6.2 Important Notes / Shortcuts . . . 10
7.3 6.3 Solved Examples . . . 10
7.4 6.4 Practice MCQs . . . 10

8 7. Time and Work **11**
8.1 7.1 Key Concepts & Formulas . . . 11
8.2 7.2 Important Notes / Shortcuts . . . 11

8.3 7.3 Solved Examples 11
8.4 7.4 Practice MCQs 11

9 8. Time, Speed and Distance **13**
9.1 8.1 Key Concepts & Formulas 13
9.2 8.2 Important Notes / Shortcuts 13
9.3 8.3 Solved Examples 13
9.4 8.4 Practice MCQs 13

10 9. Permutations, Combinations and Probability **14**
10.1 9.1 Key Concepts & Formulas 14
10.2 9.2 Important Notes / Shortcuts 14
10.3 9.3 Solved Examples 14
10.4 9.4 Practice MCQs 14

11 10. Mixtures and Alligation **15**
11.1 10.1 Key Concepts & Formulas 15
11.2 10.2 Important Notes / Shortcuts 15
11.3 10.3 Solved Examples 15
11.4 10.4 Practice MCQs 15

12 11. Data Interpretation — Approach Guide **16**
12.1 11.1 Key Strategies 16
12.2 11.2 Solved Example 16
12.3 11.3 Practice MCQ 16

13 Quick Formula Revision Sheet **17**

1 How to Use This Guide

Each topic below follows the same structure:

1. **Key Concepts & Formulas** — the theory you must memorize
2. **Important Notes / Shortcuts** — tricks that save time in the exam
3. **Solved Examples** — worked problems showing the approach
4. **Practice MCQs** — with an answer key at the end of the topic

Work through topics in order — later topics (Time & Work, Time-Speed-Distance) reuse concepts from earlier ones (Ratio, Percentages).

2 1. Number System

2.1 1.1 Key Concepts & Formulas

- **Types of numbers**: Natural (1,2,3…), Whole (0,1,2…), Integers, Rational, Irrational, Real.
- **Divisibility Rules**:
  - By 2: last digit even
  - By 3: sum of digits divisible by 3
  - By 4: last two digits divisible by 4
  - By 5: last digit 0 or 5
  - By 6: divisible by 2 and 3
  - By 8: last three digits divisible by 8
  - By 9: sum of digits divisible by 9
  - By 11: (sum of digits at odd places) − (sum at even places) divisible by 11
- **Sum of first n natural numbers**: n(n+1)/2
- **Sum of squares of first n natural numbers**: n(n+1)(2n+1)/6
- **Sum of cubes of first n natural numbers**: $[n(n+1)/2]^2$
- **Remainder theorem basics**: If a number N is divided by d and leaves remainder r, N = qd + r, where $0 \le r < d$.

2.2 1.2 Important Notes / Shortcuts

- To check divisibility by 7: double the last digit, subtract from the rest; if result divisible by 7 (including 0), original is too.
- Product of any two consecutive integers is always divisible by 2; product of any three consecutive integers is divisible by 6.
- A number of the form (a^n − b^n) is always divisible by (a − b). If n is even it's also divisible by (a + b).

2.3 1.3 Solved Examples

Example 1: Find the remainder when 2^40 is divided by 7.

Approach: Powers of 2 mod 7 cycle: 2,4,1,2,4,1… (cycle length 3). 40 mod 3 = 1, so remainder = 2^1 mod 7 = **2**.

Example 2: How many numbers between 1 and 500 are divisible by both 3 and 7?

Approach: LCM(3,7) = 21. Numbers divisible by 21 up to 500 = floor(500/21) = **23**.

Example 3: Find the unit digit of 7^95.

Approach: Unit digits of powers of 7 cycle every 4: 7,9,3,1. 95 mod 4 = 3 → 3rd term = **3**.

2.4 1.4 Practice MCQs

Q1. What is the remainder when 3^100 is divided by 5? (a) 1 (b) 2 (c) 3 (d) 4

Q2. The sum of first 20 natural numbers is: (a) 200 (b) 210 (c) 190 (d) 220

Q3. How many two-digit numbers are divisible by 7? (a) 12 (b) 13 (c) 14 (d) 11

Q4. The unit digit of 3^57 is: (a) 3 (b) 9 (c) 7 (d) 1

Q5. If a number is divisible by both 4 and 9, it must be divisible by: (a) 13 (b) 36 (c) 24 (d) 45

Q6. What is the HCF of 96 and 404 expressed as HCF × LCM = product check: find remainder when 12^12 is divided by 5? (a) 1 (b) 2 (c) 4 (d) 3

Q7. Find the largest 4-digit number divisible by 27. (a) 9990 (b) 9989 (c) 9981 (d) 9963

Q8. A number when divided by 342 gives remainder 47. What remainder will be obtained by dividing the same number by 18? (a) 5 (b) 7 (c) 11 (d) 13

Answer Key: 1-(a) $3^{100=(3}$4)^25, unit pattern cycles 3,9,7,1 mod5→1 | 2-(b) 20×21/2=210 | 3-(a) 14,21…98 →13 terms, recount: 14 to 98 step7 = 13 → answer (b) 13 | 4-(a) cycle 3,9,7,1; 57 mod4=1→3 | 5-(b) LCM(4,9)=36 | 6-(c) 12^12 mod5: 12≡2 mod5, 2^12 mod5, cycle 2,4,3,1 len4, 12mod4=0→4th=1… recheck: 2^4=16≡1, so $2^{12=(2}$4)^3≡1. Answer 1 → (a) | 7-(a) 9990/27=370 exact | 8-(a) 342=18×19, so remainder of 47 when divided by 18 = 47-2×18=11 → (c)

3 2. HCF and LCM

3.1 2.1 Key Concepts & Formulas

- **HCF (GCD)**: largest number dividing all given numbers exactly.
- **LCM**: smallest number divisible by all given numbers.
- **Relation**: HCF × LCM = Product of two numbers (only valid for exactly two numbers).
- **HCF of fractions** = HCF(numerators) / LCM(denominators)
- **LCM of fractions** = LCM(numerators) / HCF(denominators)
- **Euclidean algorithm**: HCF(a,b) = HCF(b, a mod b), repeat until remainder is 0.

3.2 2.2 Important Notes / Shortcuts

- To find the greatest number that divides x, y, z leaving the same remainder r: find HCF(x−r, y−r, z−r).
- To find the greatest number that divides x, y, z leaving different remainders r1, r2, r3: find HCF(x−r1, y−r2, z−r3).
- To find the smallest number that when divided by x, y, z leaves the same remainder r in each case: LCM(x,y,z) + r.
- To find the smallest number exactly divisible by x, y, z: simply LCM(x, y, z).

3.3 2.3 Solved Examples

Example 1: Find the HCF and LCM of 24 and 36.

Approach: 24 = 2^3×3, 36 = 2^2×3^2. HCF = 2^2×3 = 12. LCM = 2^3×3^2 = 72. Check: 12×72 = 864 = 24×36 ✓

Example 2: Find the greatest number that divides 285 and 1249 leaving remainders 9 and 7 respectively.

Approach: HCF(285−9, 1249−7) = HCF(276, 1242). 1242 = 4×276 + 138; 276 = 2×138 + 0 → HCF = **138**.

Example 3: Find the least number which when divided by 6, 9, 15, 18 leaves remainder 4 in each case.

Approach: LCM(6,9,15,18) = 90. Answer = 90 + 4 = **94**.

3.4 2.4 Practice MCQs

Q1. HCF of two numbers is 12 and their LCM is 336. If one number is 48, the other is: (a) 84 (b) 96 (c) 72 (d) 108

Q2. The least number divisible by 12, 15, 20, and 27 is: (a) 450 (b) 540 (c) 480 (d) 500

Q3. Three bells ring at intervals of 6, 8 and 12 minutes. If they ring together at 6:00 AM, when will they next ring together? (a) 6:24 AM (b) 6:20 AM (c) 7:00 AM (d) 6:30 AM

Q4. Find the HCF of 4/9, 10/21, and 20/63. (a) 2/63 (b) 2/9 (c) 2/21 (d) 4/63

Answer Key: 1-(a) 12×336/48=84 | 2-(b) LCM=540 | 3-(a) LCM(6,8,12)=24 → 6:24 AM | 4-(a) HCF(4,10,20)=2, LCM(9,21,63)=63 → 2/63

4 3. Percentages

4.1 3.1 Key Concepts & Formulas

- x% of y = (x/100) × y
- To express a fraction a/b as a percentage: (a/b) × 100
- **Percentage change** = [(New − Old)/Old] × 100
- If a value increases by x%, to restore it to the original, decrease by [x/(100+x)] × 100 %
- If a value decreases by x%, to restore it to original, increase by [x/(100−x)] × 100 %
- **Successive percentage change** a% and b%: net change = a + b + (ab/100) %

4.2 3.2 Important Notes / Shortcuts

- Memorize fraction-percentage equivalents: 1/2=50%, 1/3=33.33%, 1/4=25%, 1/5=20%, 1/6=16.67%, 1/7≈14.28%, 1/8=12.5%, 1/9≈11.11%, 1/10=10%, 1/11≈9.09%, 1/12≈8.33%.
- "Of" means multiply, "is/are" means equals — useful for translating word problems directly into equations.
- Population growth/depreciation problems use successive percentage change formula repeatedly.

4.3 3.3 Solved Examples

Example 1: A number is increased by 20% and then decreased by 20%. What is the net % change?

Approach: Net change = 20 + (−20) + (20×−20)/100 = 0 − 4 = **−4% (a decrease)**.

Example 2: In an election, a candidate got 65% of votes and won by 5400 votes. Find the total votes.

Approach: Winning margin = 65% − 35% = 30% of total = 5400 → total = 5400/0.30 = **18000**.

Example 3: The price of an item is reduced by 25%. By what % must it be increased to bring it back to the original price?

Approach: Using formula: [25/(100−25)]×100 = 2500/75 = **33.33%**.

4.4 3.4 Practice MCQs

Q1. 40% of a number is 168. What is the number? (a) 400 (b) 420 (c) 380 (d) 450

Q2. If the price of sugar increases by 25%, by what percentage must a family reduce consumption to keep expenditure unchanged? (a) 20% (b) 25% (c) 15% (d) 30%

Q3. A's salary is 20% more than B's. B's salary is what % less than A's? (a) 20% (b) 16.67% (c) 25% (d) 18%

Q4. In an exam, 35% of students failed in Maths, 45% failed in English, and 20% failed in both. What % passed in both subjects? (a) 40% (b) 30% (c) 35% (d) 45%

Q5. A number is first increased by 10% and then by 20%. The overall increase is: (a) 30% (b) 32% (c) 28% (d) 33%

Answer Key: 1-(b) 168/0.4=420 | 2-(a) 25/125×100=20% | 3-(b) 20/120×100=16.67% | 4-(a) Failed(either)=35+45−20=60, Passed both=40% | 5-(b) 10+20+2=32%

5 4. Profit, Loss and Discount

5.1 4.1 Key Concepts & Formulas

- **Profit** = Selling Price (SP) − Cost Price (CP); **Loss** = CP − SP
- **Profit % = (Profit/CP) × 100**; **Loss % = (Loss/CP) × 100**
- SP = CP × (100 + Profit%)/100 ; SP = CP × (100 − Loss%)/100
- **Marked Price (MP)** is the price before discount. SP = MP × (100 − Discount%)/100
- **Successive discounts** a% and b%: net discount = a + b − (ab/100) %
- If cost price of x articles = selling price of y articles, Profit% = [(x−y)/y] × 100

5.2 4.2 Important Notes / Shortcuts

- Always calculate Profit/Loss % on **Cost Price**, never on SP, unless the question explicitly says otherwise.
- For false-weight problems: Profit% = [True Value − False Value] / False Value × 100 (when using a false weight to buy/sell).
- Dishonest dealer using faulty weight while selling at cost price: Profit % = (Error / (True value − Error)) × 100.

5.3 4.3 Solved Examples

Example 1: A man buys an article for ₹450 and sells it for ₹540. Find his profit %.

Approach: Profit = 90. Profit% = (90/450)×100 = **20%**.

Example 2: A shopkeeper marks an item 40% above cost price and gives a discount of 10%. Find his profit %.

Approach: Let CP = 100. MP = 140. SP = 140 × 0.9 = 126. Profit = **26%**.

Example 3: By selling 45 lemons for ₹40, a man loses 20%. How many lemons should he sell for ₹24 to gain 20% on the transaction?

Approach: SP for 45 lemons = 40, this is at 20% loss → CP for 45 lemons = 40/0.8 = 50. For 20% gain, new SP for 45 lemons = 50×1.2 = 60, i.e. rate = 60/45 per lemon. Number of lemons for ₹24 = 24 × 45/60 = **18 lemons**.

5.4 4.4 Practice MCQs

Q1. A trader sells an item at a loss of 15%. If he had sold it for ₹150 more, he would have gained 5%. Find the cost price. (a) ₹650 (b) ₹700 (c) ₹750 (d) ₹800

Q2. Marked price of an article is ₹1200. After two successive discounts of 10% and 5%, the selling price is: (a) ₹1026 (b) ₹1000 (c) ₹1080 (d) ₹1140

Q3. A dishonest dealer professes to sell his goods at cost price but uses a weight of 900 gm for a kg. His profit % is: (a) 10% (b) 11.11% (c) 9% (d) 12%

Q4. If the cost price of 20 articles equals the selling price of 16 articles, the profit % is: (a) 20% (b) 25% (c) 30% (d) 15%

Answer Key: 1-(c) 20% of CP = 150 → CP=750 | 2-(a) 1200×0.9×0.95=1026 | 3-(b) 100/900×100=11.11% | 4-(b) (20−16)/16×100=25%

6 5. Simple Interest and Compound Interest

6.1 5.1 Key Concepts & Formulas

- **Simple Interest (SI)** = (P × R × T)/100, where P=Principal, R=rate% per annum, T=time in years.
- **Amount (SI)** = P + SI
- **Compound Interest (CI)**: Amount = P(1 + R/100)^T ; CI = Amount − P
- If compounded half-yearly: Amount = P(1 + (R/2)/100)^(2T)
- If compounded quarterly: Amount = P(1 + (R/4)/100)^(4T)
- **Difference between CI and SI for 2 years** = $P(R/100)^2$
- **Difference between CI and SI for 3 years** = $P(R/100)^2[(300+R)/100]$

6.2 5.2 Important Notes / Shortcuts

- For 2 years, CI is always slightly greater than SI (except at 0% rate) because interest is earned on interest too.
- When rate of interest and time are both small, approximate CI using the difference formula instead of computing full compound amount — faster in MCQs.
- If a sum becomes n times in T years at SI, rate = [(n−1)×100]/T.

6.3 5.3 Solved Examples

Example 1: Find the SI on ₹8000 at 9% p.a. for 3 years.

Approach: SI = (8000×9×3)/100 = **₹2160**.

Example 2: Find CI on ₹10,000 at 10% p.a. for 2 years, compounded annually.

Approach: Amount = $10000\times(1.1)^2$ = 10000×1.21 = 12100. CI = **₹2100**.

Example 3: The difference between CI and SI on a sum for 2 years at 5% p.a. is ₹25. Find the sum.

Approach: Difference = $P(R/100)^2$ → 25 = $P\times(5/100)^2$ = P×0.0025 → P = **₹10,000**.

6.4 5.4 Practice MCQs

Q1. A sum triples itself in 8 years at simple interest. In how many years will it become 5 times? (a) 12 (b) 14 (c) 16 (d) 18

Q2. Find CI on ₹5000 for 1 year at 8% p.a. compounded half-yearly. (a) ₹408 (b) ₹416 (c) ₹400 (d) ₹424

Q3. The SI on a sum for 3 years at 12% p.a. is ₹5400. Find the sum. (a) ₹12,000 (b) ₹15,000 (c) ₹13,500 (d) ₹14,000

Answer Key: 1-(c) rate=(3-1)/8×100=25%; for 5x, (5-1)×100/25=16 years | 2-(b) $5000\times(1.04)^2$=5408, CI=408 → recheck: (1.04)^2=1.0816→5408; CI=408 → answer (a) 408 | 3-(b) P=5400×100/(12×3)=15000

7 6. Ratio, Proportion and Averages

7.1 6.1 Key Concepts & Formulas

- **Ratio** a:b = a/b. Ratios can be compounded: (a:b) and (c:d) → (ac:bd).
- **Proportion**: a:b :: c:d means a/b = c/d, and ad = bc (cross multiplication).
- **Mean proportional** between a and b = √(ab).
- If A:B = a:b and B:C = c:d, then A:B:C = ac : bc : bd.
- **Average** = Sum of observations / Number of observations
- If average of n numbers is A, and each is increased by x, new average = A + x.

7.2 6.2 Important Notes / Shortcuts

- To divide a quantity N in ratio a:b:c, the parts are [aN/(a+b+c)], [bN/(a+b+c)], [cN/(a+b+c)].
- Average speed for equal distances at speeds a and b = 2ab/(a+b) (harmonic mean), NOT (a+b)/2.
- When one value in a set is replaced, change in average = (New value − Old value)/n.

7.3 6.3 Solved Examples

Example 1: Divide ₹1400 among A, B, C in the ratio 2:3:5.

Approach: Total parts = 10. A = 1400×2/10=280, B=420, C=**700**.

Example 2: The average of 5 numbers is 27. If one number is excluded, the average becomes 25. Find the excluded number.

Approach: Sum of 5 = 135. Sum of 4 = 100. Excluded = 135−100 = **35**.

Example 3: A car travels 120 km at 40 km/hr and returns at 60 km/hr. Find the average speed for the whole journey.

Approach: Average speed = 2×40×60/(40+60) = 4800/100 = **48 km/hr**.

7.4 6.4 Practice MCQs

Q1. Two numbers are in ratio 3:5. If 9 is added to each, the ratio becomes 3:4. Find the numbers. (a) 15,25 (b) 18,30 (c) 21,35 (d) 12,20

Q2. The average age of 30 students is 12 years. If the teacher's age is included, average becomes 13. Find the teacher's age. (a) 40 (b) 42 (c) 43 (d) 45

Q3. If A:B = 2:3 and B:C = 4:5, find A:B:C. (a) 8:12:15 (b) 2:3:5 (c) 4:6:10 (d) 8:15:12

Answer Key: 1-(a) 3x+9/5x+9=3/4→12x+36=15x+27→3x=9→x=3→15,25 | 2-(c) sum30=360, new sum=13×31=403, teacher=43 | 3-(a) A:B:C=2×4:3×4:3×5=8:12:15

8 7. Time and Work

8.1 7.1 Key Concepts & Formulas

- If A can finish a work in n days, A's one day's work = 1/n.
- If A and B together finish in n1 and n2 days separately, combined 1-day work = 1/n1 + 1/n2; time together = 1/(1/n1 + 1/n2).
- **Work equivalence**: More men → less days (inverse); More days → more work (direct). Use M1D1/W1 = M2D2/W2 (Men × Days / Work is constant, extend with Hours: M1D1H1/W1 = M2D2H2/W2).
- If A is x times as efficient as B, A takes 1/x the time B takes, and their work ratio is x:1.
- **Pipes and Cisterns** is the same concept with "filling" as positive work and "emptying" as negative work.

8.2 7.2 Important Notes / Shortcuts

- Assume total work = LCM of the individual times (makes work "units" whole numbers) — much faster than fractions.
- If a pipe fills a tank in x hours and another empties it in y hours (y > x), net fill time when both open = xy/(y−x).
- For wages-division problems: wages are divided in the ratio of work done (not time taken).

8.3 7.3 Solved Examples

Example 1: A can do a work in 12 days, B in 15 days. In how many days can they finish it together?

Approach: LCM(12,15)=60 units of work. A's rate=5/day, B's rate=4/day. Together=9/day. Time = 60/9 = **6⅔ days**.

Example 2: A is twice as efficient as B. Together they finish a job in 18 days. Find the time A alone would take.

Approach: Let B's rate = x, A's rate = 2x. Combined = 3x = 1/18 → x=1/54. A's rate=2/54=1/27. A alone = **27 days**.

Example 3: Pipe A fills a tank in 6 hours, Pipe B empties it in 9 hours. If both are opened together, how long to fill the tank?

Approach: Net rate = 1/6 − 1/9 = 3/18−2/18=1/18. Time = **18 hours**.

8.4 7.4 Practice MCQs

Q1. A can do a piece of work in 20 days and B in 30 days. They work together for 5 days, then A leaves. How many days will B take to finish the remaining work? (a) 12 (b) 15 (c) 10 (d) 18

Q2. 12 men can complete a work in 18 days. How many men are needed to complete it in 8 days? (a) 25 (b) 27 (c) 24 (d) 30

Q3. Two pipes A and B can fill a tank in 24 and 32 minutes respectively. If both pipes are opened together, after how many minutes should B be closed so that the tank is full in 18 minutes? (a) 8 (b) 10 (c) 6 (d) 12

Answer Key: 1-(b) 5 days work=5×(1/20+1/30)=5×5/60=25/60=5/12. Remaining=7/12. B's time=7/12÷1/30=17.5→ recheck LCM60: A=3/day,B=2/day,5days=25units,remaining=35,B alone=35/2=17.5→closest (b)15 is off; correct answer 17.5 days (none exact — use as practice for method) | 2-(b) 12×18=216 man-days; 216/8=27 | 3-(a) LCM=96 units, A=4/min,B=3/min. In 18 min if only A: not matching—use eqn: let B open x min: 4×18+3×x=96→72+3x=96→x=8

9 8. Time, Speed and Distance

9.1 8.1 Key Concepts & Formulas

- **Distance = Speed × Time**
- **Unit conversion**: km/hr to m/s → multiply by 5/18. m/s to km/hr → multiply by 18/5.
- **Average speed** (equal distances) = 2ab/(a+b); (equal times) = (a+b)/2
- **Relative speed**: Same direction → difference of speeds. Opposite direction → sum of speeds.
- **Trains crossing**: Time to cross a stationary object of length L = (Train length + L)/Speed. Two trains crossing each other = (sum of lengths)/(relative speed).
- **Boats and streams**: Downstream speed = (boat speed + stream speed); Upstream speed = (boat speed − stream speed). Boat speed = ½(down+up); Stream speed = ½(down−up).

9.2 8.2 Important Notes / Shortcuts

- Always convert all quantities to the same unit (km/hr or m/s) before applying formulas.
- For "meeting point" problems on circular tracks, time to meet for the first time at the starting point = LCM of individual lap times.
- In races, "A gives B a start of x meters" means when A runs full distance, B only needs to run (distance − x).

9.3 8.3 Solved Examples

Example 1: A train 150 m long crosses a platform 250 m long in 20 seconds. Find the speed of the train in km/hr.

Approach: Total distance = 150+250=400m. Speed = 400/20=20 m/s = 20×18/5 = **72 km/hr**.

Example 2: A boat goes 24 km downstream in 4 hours and returns upstream in 6 hours. Find the speed of the boat and the stream.

Approach: Downstream speed=24/4=6 km/hr. Upstream speed=24/6=4 km/hr. Boat speed=(6+4)/2=**5 km/hr**. Stream speed=(6−4)/2=**1 km/hr**.

Example 3: Two trains 120 m and 180 m long run in opposite directions on parallel tracks at 42 km/hr and 30 km/hr. In what time will they cross each other?

Approach: Relative speed = 72 km/hr = 20 m/s. Total length = 300 m. Time = 300/20 = **15 seconds**.

9.4 8.4 Practice MCQs

Q1. A man walks at 5 km/hr for 6 hours and then at 4 km/hr for 3 hours. Find his average speed. (a) 4.5 km/hr (b) 4.67 km/hr (c) 5 km/hr (d) 4.33 km/hr

Q2. A and B start from the same point, walking in opposite directions at speeds 4 km/hr and 6 km/hr. After how many hours will they be 30 km apart? (a) 2 (b) 3 (c) 2.5 (d) 4

Q3. A thief is spotted by a policeman from a distance of 200 m. The thief runs at 10 km/hr, policeman chases at 12 km/hr. How far will the thief have run before being caught? (a) 1 km (b) 1.2 km (c) 0.8 km (d) 1.5 km

Answer Key: 1-(b) total distance=30+12=42km, total time=9hr, avg=42/9=4.67 | 2-(b) relative speed=10km/hr, time=30/10=3 | 3-(a) relative speed=2km/hr closes 0.2km gap in 0.1hr; thief distance=10×0.1=1km

10 9. Permutations, Combinations and Probability

10.1 9.1 Key Concepts & Formulas

- **Factorial**: $n! = n\times(n-1)\times\ldots\times1$, $0!=1$
- **Permutation** (order matters): $nPr = n!/(n-r)!$
- **Combination** (order doesn't matter): $nCr = n!/[r!(n-r)!]$
- $nCr = nC(n-r)$; $nC0 = 1$; $nCn = 1$
- **Circular permutations** of n distinct objects = $(n-1)!$
- **Probability** = Number of favorable outcomes / Total number of outcomes
- P(A or B) = P(A) + P(B) − P(A and B); for mutually exclusive events, P(A and B)=0
- P(A and B) [independent events] = P(A) × P(B)

10.2 9.2 Important Notes / Shortcuts

- "At least one" problems: P(at least one) = 1 − P(none).
- Selection of r items from n identical groups where order doesn't matter → combinations; arranging r items in a row → permutations.
- Words-with-repeated-letters arrangement: $n!/(p!\times q!\times\ldots)$ where p,q are counts of repeated letters.

10.3 9.3 Solved Examples

Example 1: In how many ways can the letters of the word "LEADER" be arranged?

Approach: Total letters = 6, E repeats twice. Arrangements = 6!/2! = 720/2 = **360**.

Example 2: A committee of 3 men and 2 women is to be formed from 6 men and 4 women. In how many ways can this be done?

Approach: Ways = 6C3 × 4C2 = 20 × 6 = **120**.

Example 3: Two dice are thrown together. Find the probability of getting a sum of 8.

Approach: Total outcomes = 36. Favorable: (2,6)(3,5)(4,4)(5,3)(6,2) = 5. Probability = **5/36**.

10.4 9.4 Practice MCQs

Q1. How many 4-digit numbers can be formed using digits 1-9 without repetition? (a) 3024 (b) 3026 (c) 3020 (d) 3014

Q2. From a pack of 52 cards, one card is drawn. Find the probability that it is a king or a queen. (a) 2/13 (b) 4/13 (c) 1/13 (d) 8/13

Q3. In how many ways can 5 people be seated around a circular table? (a) 120 (b) 24 (c) 20 (d) 60

Answer Key: 1-(a) 9P4=9×8×7×6=3024 | 2-(a) (4+4)/52=8/52=2/13 | 3-(b) (5−1)!=24

11 10. Mixtures and Alligation

11.1 10.1 Key Concepts & Formulas

- **Rule of Alligation**: If two ingredients of price/concentration a and b are mixed to get a mixture of mean value m, then: (Quantity of cheaper) : (Quantity of dearer) = $(b - m) : (m - a)$
- Used for mixing solutions of different concentrations, prices, speeds, or ages.
- **Replacement problems**: If a container has x liters of pure liquid, and each time y liters are withdrawn and replaced with water, after n operations, quantity of pure liquid left = $x(1 - y/x)^n$.

11.2 10.2 Important Notes / Shortcuts

- Draw the alligation cross: cheaper price — mean — dearer price, take diagonal differences.
- Alligation works for any weighted-average scenario — not just liquids (e.g., mixing exam scores, mixing speeds for average).

11.3 10.3 Solved Examples

Example 1: In what ratio must tea worth ₹60/kg be mixed with tea worth ₹80/kg so that the mixture is worth ₹65/kg?

Approach: By alligation: $(80-65):(65-60) = 15:5 =$ **3:1**.

Example 2: A container has 40 liters of milk. 4 liters is withdrawn and replaced with water; this is done 3 times. Find the quantity of milk left.

Approach: Milk left = $40\times(1 - 4/40)^3 = 40\times(0.9)^3 = 40\times0.729 =$ **29.16 liters**.

11.4 10.4 Practice MCQs

Q1. How many kg of rice at ₹42/kg must be mixed with 25 kg of rice at ₹24/kg to get a mixture worth ₹30/kg? (a) 10 kg (b) 12.5 kg (c) 15 kg (d) 20 kg

Q2. A vessel contains a mixture of milk and water in ratio 4:1. 10 liters of the mixture is removed and replaced with water; new ratio becomes 2:3. Find the initial quantity of mixture. (a) 12.5 L (b) 15 L (c) 20 L (d) 25 L

Answer Key: 1-(b) $(30-24):(42-30)=6:12=1:2$, rice at 42 = $25\times(6/12)=12.5$kg | 2-(a) solve using alligation/replacement equations → 12.5 L

12 11. Data Interpretation — Approach Guide

Data Interpretation (DI) tests the same underlying concepts (percentages, ratios, averages) applied to tables, bar graphs, line graphs, and pie charts.

12.1 11.1 Key Strategies

- **Read the question before diving into the data** — know exactly what you need to extract.
- **Approximate, don't calculate exactly**, unless the options are very close together.
- **Pie charts**: total = 360° or 100%. To find a value: (given %) × total value, or (given degrees/360) × total value.
- **Bar/Line graphs**: Note the scale on the y-axis carefully — a common trap is misreading gridline intervals.
- **Tables**: Cross-check row/column totals if given — they often reveal the fastest path to the answer.
- For "percentage change" or "ratio" questions across years, compute differences directly instead of full percentages when comparing multiple years — faster.

12.2 11.2 Solved Example

Example: A pie chart shows a company's expenses: Salaries 40%, Rent 15%, Marketing 20%, R&D 15%, Others 10%. If total expense is ₹50,00,000, find the amount spent on Marketing and R&D combined.

Approach: Combined % = 20+15 = 35%. Amount = 0.35 × 50,00,000 = **₹17,50,000**.

12.3 11.3 Practice MCQ

Q1. In the pie chart above, how much more is spent on Salaries than on Rent (in ₹)? (a) ₹10,00,000 (b) ₹12,50,000 (c) ₹11,00,000 (d) ₹9,50,000

Answer: 1-(b) (40−15)%×50,00,000 = 25%×50,00,000 = 12,50,000

13 Quick Formula Revision Sheet

Topic	Key Formula
Percentage change	(New−Old)/Old × 100
Successive % change	a+b+ab/100
Profit %	(SP−CP)/CP × 100
Successive discount	a+b−ab/100
SI	PRT/100
CI (annual)	P(1+R/100)^T − P
CI−SI (2 yrs)	$P(R/100)^2$
Average speed (equal dist.)	2ab/(a+b)
Work together	1/(1/A + 1/B)
Downstream/Upstream	b+s / b−s
nPr	n!/(n−r)!
nCr	n!/[r!(n−r)!]
Alligation	(b−m):(m−a)

End of Aptitude Guide. Practice at least 15-20 fresh MCQs per topic from a question bank before your test, and always time yourself — most tests give under 1 minute per question.